

Chronological Synthesis of the Spherical Substrate Line

Executive Summary

The newest PDF does not merely update the archive; it changes the interpretive rule for the whole corpus. Its core instruction is to stop judging the archive as if it were fundamentally a proposal about rasterization, ray marching, voxel storage, or a conventional frame loop. Instead, it argues that those are downstream projection tools, while the actual proposal is an ontology of directly queryable relations, events, phases, and identities. In that reading, the archive's "cone," "SDF = 0," "one-bit," "double vacuum," and "hourglass" motifs are not primarily graphics primitives. They are operators in a shared state space. That is the decisive shift in the corpus. [filecite\turn0file0\20-L20-L33](#)

Read chronologically, the archive moves through a consistent arc. It begins with reaction-diffusion balance, collapse, gradients, and distributed state; then introduces cone and polar language; then develops a layered shared ontology for geometry, sensing, uncertainty, and semantics; then returns obsessively to boundary, parity, persistence, and continuity-through-zero; and finally, in the 19 July documents, recasts the whole project as a non-raster equation-world built from compatibility, event solving, transition rules, lineage, and an irreducible event log. The latest PDF itself summarizes that progression explicitly and treats its own contribution as a constructive correction: formalize the concept on its own terms before comparing it to established engines or renderers. [filecite\turn0file0\54-L76](#) [filecite\turn0file0\593-L604](#)

The strongest unified concept is therefore **not** "replace all world structure with spheres" and **not** "render the world by spherical raymarching." It is a more specific claim: a finite grammar can generate a continuous, phase-layered, event-driven state manifold in which local spherical or radial-angular supports organize relevance, coupling, and measurement, while identities are preserved by lineage and invariants rather than by instantaneous coordinates. In that concept, "spherical" is best understood as the local substrate of support, shelling, orientation, and radial reach, not as a universal replacement for every representation in the system. This interpretation matches the archive's own mid-July corrections, which repeatedly reject both "rasterized-everything" and "polar-everything" in favor of a layered or query-driven ontology. [filecite\turn0file0\95-L118](#) [filecite\turn0file0\258-L301](#) (Polar Coordinates as the Missing Local Chart for the Aerosol Cone Architecture.pdf, pp. 1–3) (Unified Architecture for Aerosol Cone Sensing, SDFs, Nested Shells, and Spherical Kinematics.pdf, pp. 1–2) (Hybrid Ontology for Unity 6.3.pdf, pp. 1–2)

On feasibility, the corpus is internally strongest when narrowed. The latest PDF is explicit that the most defensible form is a restricted substrate that can answer state-at-time and next-event queries without universal frame stepping, but only for controlled relation families. It also explicitly demotes the grander claims—universal $O(1)$, zero memory, world-equals-general-AI—from doctrine to research hypotheses. That is the right reading of the corpus. Under that narrowed interpretation, the "spherical substrate line" is feasible as a prototype-worthy computational medium. As a universal replacement for graphics, simulation, memory, cognition, and semantics all at once, it remains unproven.

[filecite\turn0file0\40-L49](#) [filecite\turn0file0\366-L375](#) [filecite\turn0file0\579-L592](#)

A necessary assumption in the analysis below is that ambiguous corpus language should be interpreted **constructively**: “sphere,” “cone,” “double vacuum,” “Klein bottle,” and “one-bit” are read as computational variables, local charts, compatibility sectors, gluing rules, and routing bits unless the documents explicitly insist on literal physics. That constructive translation is not imposed from outside; it is exactly the move the latest PDF tells the reader to make. [filecite\turn0file0\130-133](#)
[filecite\turn0file0\123-182](#)

Chronological Corpus

The timeline below follows the chronology stated in the latest PDF’s “Chronological development” and “Source basis” pages, then tightens it with the archive’s own filenames and page references. The archive order matters because the core idea does not appear fully formed; it stabilizes through repeated corrections. [filecite\turn0file0\54-L76](#) [filecite\turn0file0\593-L604](#)

Date or cohort	Core documents in the archive	What the document family is doing	Advance in the spherical substrate line	Where it diverges or slides back
Early July	<i>Reaction-Diffusion-Collapse-Explained.pdf</i> (pp. 1–11)	Establishes balance, collapse, hysteresis, gradients, topological moats, and distributed state before object inventories	Introduces the idea that law-of-change precedes object storage; world as evolving field rather than mesh inventory	Still framed through explanatory analogies and simulation language, not yet a formal substrate
Early July	<i>posterize-shader-black-and-white-dots...aerosol-spray-in-a-cone...lighthouse.pdf</i> (pp. 1–5, 25–33)	Starts from a shader-and-display framing, then partially migrates toward compute-field language	Brings in the cone as directional support and the idea that the same field can serve both display and machine access	Repeatedly falls back into rendering vocabulary, especially posterization, dither, and screen-space discussion

Date or cohort	Core documents in the archive	What the document family is doing	Advance in the spherical substrate line	Where it diverges or slides back
Early July	<i>firefly-trapped-in-a-sphere...cartesian...polar...relationship.pdf</i> (pp. 1–5, 9–11)	Establishes radial/angular organization, manifold language, tangent spaces, and polar/spherical transformation imagery	Makes “sphere” and “polar” native for constrained local motion and support	Risks overextending coordinate transforms into metaphysical universals
Mid July shared ontology cohort	<i>Polar Coordinates as the Missing Local Chart...</i> (pp. 1–3), <i>Unified Architecture for Aerosol Cone Sensing...Spherical Kinematics.pdf</i> (pp. 1–2), <i>Hybrid Ontology for Unity 6.3.pdf</i> (pp. 1–2), <i>Feasibility Assessment of the Brainstem Concept.pdf</i> (pp. 1–2)	Consolidates the layered stack: graph/RD, SDF geometry, local cone sensing, localization, fuzzy or semantic lift	Critical correction: keep world structure hybrid or geometric, but make local sensing radial-angular; “spherical” becomes a local chart, not the whole world	Still partly anchored to Unity, TSDF/ESDF, shaders, and implementation splits
Mid July invariants and embodiment cohort	<i>Particle-Collisions-Explained-Bunches,-Not-Single-Particles.pdf</i> (pp. 1–5), <i>GlagStone-1-Bit-City-Builder-Explained (1).pdf</i> (pp. 3–8), <i>StepsWithout0.pdf</i> (pp. 22, 27, 43–45), <i>Cobalt_ The Immutable Anchor...pdf</i> (p. 1)	Develops one-bit routing, event crossing, dt=0 rhetoric, continuity-through-zero, and stable anchors	Pushes the archive toward threshold semantics, parity flips, zero-crossings, and persistent invariants	This cohort is the most metaphor-heavy and least disciplined; it often confuses visual style, psychology, and formal mechanism

Date or cohort	Core documents in the archive	What the document family is doing	Advance in the spherical substrate line	Where it diverges or slides back
Late July	<i>hollowland-double-vacuum.pdf</i> (pp. 29–35, with pivot setup on pp. 6–8)	Explodes the concept into its grandest form: ontological UUIDs, non-sieved algebraic space, hourglasses, phase masks, under-32 MB worlds, AI/world identity	Introduces the full vocabulary later normalized by the newest PDF: phase, one-bit, double vacuum, hourglass, lineage, and algebraic event persistence	Also makes the archive's least controlled claims, especially around $O(1)$, memory, and universal world/AI fusion
Late July constructive audit	<i>Feasibility and Constructive Potential Assessment of the Source Document.pdf</i> (pp. 1–7)	Separates metaphor from implementable mechanism and recommends a much narrower prototype path	Converts the archive from rhetoric into falsifiable program fragments: implicit relations, exact or conservative time-of-impact, grammar layer, low-bit state tags	Still interprets much through continuous collision detection and engine baselines

Date or cohort	Core documents in the archive	What the document family is doing	Advance in the spherical substrate line	Where it diverges or slides back
Latest attached PDF	<i>Beyond the Raster Trap Chronological Synthesis</i> (pp. 2–17)	Performs the cleanest internal synthesis: relation algebra, phase sheets, compatibility, event solving, lineage, irreducible event log, and Equation World Zero	Finalizes the through line as a declarative state-and-event calculus with local spherical support and non-raster semantics	Deliberately brackets external validation, so it establishes internal coherence better than external proof

What matters most in that sequence is the **direction of correction**. The archive first tends to imagine cones, spheres, and 1-bit transitions as graphics-adjacent or sensor-adjacent metaphors. The mid-July analyses then insist that local radial-angular organization belongs mainly to sensing and support. The 19 July synthesis completes the move by treating the core as an event calculus in which rendering, if used at all, is secondary. [filecite\turn0file0\24-L29](#) [filecite\turn0file0\97-L118](#)

Unified Spherical Concept

The unified concept can be stated compactly:

A finite grammar generates a continuous state manifold whose locally spherical or radial-angular supports define relevance, coupling, and measurement; events occur when relations cross explicit transition surfaces under compatibility constraints; identity persists by lineage and invariants rather than by coordinates; and projection into images is optional, not constitutive. [filecite\turn0file0\134-L49](#)
[filecite\turn0file0\564-L592](#)

That statement is tighter than most individual source documents, but it is faithful to the corpus once the late reframing is applied. The newest PDF defines the world as a finite grammar plus continuous relations, phase sheets, invariants, and an irreducible event log. It also translates the archive's recurring vocabulary into operational terms: **B = 0** becomes an event surface or transition relation, **cones/hourglasses** become domains of influence or admissible relevance, **phi** becomes a phase law, **one-bit** becomes routing or parity, **double vacuum** becomes phase-separated co-location, and **ontological address** becomes lineage-based identity. [filecite\turn0file0\123-L182](#)
[filecite\turn0file0\564-L583](#)

The formal core proposed in the latest PDF is the clearest mathematical condensation in the corpus:

$$Q = P^n \times \mathbb{R}_{time} \times S^1_{phase} \times \mathbb{Z}_{2sheet} \times A_{address} \times B_{branch}$$

$$\begin{aligned}
q_e(t) &= (x_e(t), t, \phi_e(t), \sigma_e(t), a_e, b_e) \\
W &= (G, R, T, I) \\
R_j(q; \theta_j) &= 0, \quad C_\alpha(q, t) \leq 0, \quad \chi(e, j, t) \in \{0, 1\} \\
t^* &= \min\{t \geq t_0 : R_j(q_e(t)) = 0, C_\alpha \leq 0, \chi = 1\} \\
q_e(t^{*+}) &= T_j(q_e(t^{*-}), \text{context})
\end{aligned}$$

In plain language, a world state is not only “where in space,” but also “when,” “in what phase,” “on which sheet,” “at which lineage-address,” and “on which branch.” Events are not pure geometry intersections; they happen only when a relation is satisfied, the state lies inside the relevant support cone, and the co-located sectors are compatible. [filecite\turn0file0\2022-L230](#)

Architecture of the equation world from the latest PDF, page 6

The architecture diagram on page 6 makes the ontology especially clear: grammar and exogenous log feed a projective-phase manifold; relations and compatibility operate inside that manifold; cone/hourglass queries select relevance; and transitions update state while preserving invariants. The same state description can then answer state-at-time, next-event, reachable-relation, identity, and agent-action queries. That is why the through line is deeper than graphics. It proposes a common substrate for physics-like transitions, sensing, persistence, and agency. [filecite\turn0file0\2022-L187-L197](#)

The specifically **spherical** part of the line needs to be stated carefully. The archive’s strongest correction, visible already in the mid-July analyses, is that spherical or polar structure belongs primarily to **local support, egocentric sensing, shell organization, radial reach, and phase coupling**—not to a global remeshing of all world state. *Polar Coordinates as the Missing Local Chart for the Aerosol Cone Architecture* and *Unified Architecture for Aerosol Cone Sensing, SDFs, Nested Shells, and Spherical Kinematics* both insist on that distinction: “keep the world model hybrid,” “make the sensor model polar,” and treat spherical coordinates as native to the local manifold of cones, shells, and bilateral kernels rather than as a total world replacement (Polar Coordinates..., pp. 1–3; Unified Architecture..., pp. 1–2). *Hybrid Ontology for Unity 6.3* makes the same point more bluntly: neither raster-everything nor polar-everything is the right reading; persistent world state, local sensing, and display should remain distinct layers (Hybrid Ontology..., pp. 1–2).

The archive’s best-developed singular notion is **double vacuum**. In the latest PDF it is given its cleanest formulation: two states can share the same coordinate x yet still fail to interact because their phase, sheet, orientation, or address are incompatible. The “second vacuum” is therefore not a stronger emptiness; it is the absence of coupling. This is one of the core places where the corpus stops being about visual metaphors and becomes a computational ontology. Same position does not imply same sector. Interaction becomes explicitly gated. [filecite\turn0file0\2022-L235-L252](#)

Operational reading of double vacuum and the quad hourglass from the latest PDF, page 8

That page 8 diagram is useful because it shows how several recurrent archive motifs collapse into one structure: the four chambers become a routing partition, the central pinch becomes a rank-changing event locus, the vertical line becomes an invariant mode or eigen-axis, and the swept arc becomes phase-controlled coupling. The image is not literal topology proof, but it is a disciplined visualization of the actual computational claim. [filecite\turn0file0\2022-L238-L243](#) [filecite\turn0file0\2022-L245-L252](#)

One more correction is crucial. In *StepsWithout0*, the archive briefly pushes toward the idea that an SDF can “go past zero” into a spherical fluid LUT and thereby bypass the frontend. But that same document later admits that without a hard zero or natural cancellation surface, the system falls back into hidden rasterization, because sampled approximation returns through the back door (*StepsWithout0.pdf*, pp. 22, 27, 43–45). The newest PDF solves that problem more cleanly: $\mathbf{B} = \mathbf{0}$ is not “keep marching through a rendered SDF”; it is the explicit transition relation where event semantics are invoked. That is a major maturation of the corpus. [filecite\turn0file0\127-137](#) [filecite\turn0file0\1213-1214](#)

The architecture can be summarized as follows:

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flowchart LR
    G[Finite grammar G] --> Q[Projective-phase state manifold Q]
    L[Exogenous log L] --> Q
    Q --> R[Relations  $R_j(q)=0$ ]
    Q --> C[Cone or hourglass support  $C(q,t)\leq 0$ ]
    Q --> X[Compatibility  $\chi$  phase/sheet/address]
    R --> E[Event time  $t^*$ ]
    C --> E
    X --> E
    E --> T[Transition  $T_j$ ]
    T --> O[Outputs: state-at-time, next-event, identity, reachable relation, agent action]

```

This is why the corpus’s mature through line is best called a **spherical-support event calculus** rather than a graphics alternative. The sphere, shell, cone, and hourglass motifs are not there to win a renderer war. They are there to define local supports, sectors, and transitions in a shared state space. [filecite\turn0file0\1258-1301](#)

Feasibility

What is realistically feasible

Under the latest PDF’s intended interpretation, the concept is strongest at the lower rungs of its own feasibility ladder: relational representation, phase-layered state, and direct state/event queries over restricted relation families. The report’s overall verdict is explicit: the architecture is conceptually coherent and prototype-feasible as a **restricted** equation-world substrate, while its scale and generality remain open research questions. That distinction should be treated as the authoritative internal feasibility judgment of the corpus. [filecite\turn0file0\1366-1375](#)

The most defensible restricted prototype is already specified as **Equation World Zero**. It uses 2D homogeneous coordinates plus continuous time, two sheets and one orientation bit, four-chamber routing at $B = 0$, a tightly bounded relation family, simple trajectory classes, finite grammar depth, lineage-based identity, and symbolic state/event outputs rather than image-first ambitions. That is a healthy prototype design because it puts the burden where the concept actually lives: state evaluation, compatibility, event solving, routing, and persistence. [filecite\turn0file0\1477-1503](#)

Computational complexity

The latest PDF offers a narrow and useful reinterpretation of the archive's $O(1)$ rhetoric. It does **not** claim universal constant cost. It claims only that, for a fixed closed expression, the cost of asking for state at time t can be independent of the number of skipped time steps. Similarly, next-event cost is dominated by compatibility checks, event solving, and branch resolution, not by frame replay. That is a strong claim, but it is much weaker than “constant-cost universe,” and much more plausible.

[filecite turns file L224-L230](#) [filecite turns file L445-L449](#)

A practical cost picture, using the corpus's own terms, looks like this:

Query or operation	Dominant cost under the intended model	Why it can beat frame stepping	Why it can still fail
<code>state_at(t)</code>	Evaluate closed state expression $q_e(t)$	Cost can be horizon-independent for fixed expressions	Expression size can grow with grammar depth or branch history
<code>next_event</code>	Compatibility + root/event solve + branch resolution	Avoids replaying all intermediate states	Event density, multiple roots, and degeneracies can explode
Cone query	Support pruning + relation evaluation inside support	Relevance is localized analytically, not by rendering an image	Poor compatibility filters turn cone queries into broad scans
Persistence rebuild	Seed + grammar + external event log	Closed dynamics can be recomputed instead of stored frame-by-frame	Exogenous novelty still accumulates and must be logged
Identity reconstruction	Generative address + invariant history	Avoids relying on current coordinate as unique identity	Splits, merges, and reconciliations need explicit lineage rules

This table is not imported from outside. It is a direct synthesis of the formal core, the “precise meaning of $O(1)$,” the claim-by-claim assessment, and the evaluation program in the latest PDF.

[filecite turns file L202-L230](#) [filecite turns file L380-L450](#) [filecite turns file L507-L559](#)

The key feasibility criterion on page 14 is especially important: the architecture only succeeds if the cost of compatibility plus event solving grows more slowly than the amount of world structure the system avoids explicitly materializing. That is the real computational test. Not whether it “sounds unlike rasterization,” but whether its pruning and solvability dominate its combinatorics.

[filecite turns file L455-L470](#)

Numerical stability

The newest PDF is unusually candid about numerical fragility. It flags long time horizons, log coordinates, irrational phase schedules, near-singular operators, tangencies, coincident surfaces, and rank changes as actual engineering risks. That list is not incidental; it tells us exactly where the prototype can fail while still being internally faithful to the concept. If the substrate relies on phase,

singular centers, and event surfaces, then conditioning is not a side issue. It is part of the ontology. [filecite turn0file0L463-L466](#)

This is where the corpus’s mature form is notably better than its earlier rhetoric. Earlier material often implied that spherical organization or 1-bit transitions would automatically bypass computational hardness. The latest PDF rejects that magical reading. It explicitly allows interval or set-valued answers, certified numerical root processes, and pragmatic uncertainty semantics. That is the right stance for a real prototype. [filecite turn0file0L221-L230](#) [filecite turn0file0L289-L297](#)

Hardware implications

The corpus now supports two implementation pathways, and they should not be confused.

The first is the **query-first headless path** favored by the latest PDF. Start from algebra, compatibility, events, transitions, identities, and logs. Do not begin by rebuilding a conventional scene or image pipeline and then claiming victory if it looks clever. The latest PDF is direct that “success is not a picture,” and that visual fidelity cannot define the prototype. [filecite turn0file0L492-L500](#) [filecite turn0file0L507-L559](#) [filecite turn0file0L588-L592](#)

The second is the **hybrid engine path** developed in the mid-July archive analyses. Those documents consistently recommend a split in which authoritative world geometry and belief remain in a stable geometric or graph-backed core, while cone integration, display-focused volumetrics, or optional raymarching live in a secondary compute stage. This is a feasible engineering bridge, but it is already a concession to conventional engine structure. It is useful as a bootstrap path, not as the purest statement of the idea (Hybrid Ontology for Unity 6.3.pdf, pp. 1–2; Aerosol Cone Vision in the Unified Engine.pdf, pp. 1–2; Aerosol Volumetrics Inside a Unified Engine Architecture.pdf, pp. 1–2).

The hardware lesson is therefore simple: if the goal is to test the **core principles**, use the headless algebraic path first. If the goal is to productize a stepping stone inside a game engine, the hybrid path is the safer bridge. The corpus supports both, but it clearly treats the first as the truer experiment. [filecite turn0file0L477-L503](#) [filecite turn0file0L544-L559](#)

Comparison to Rasterization, Raymarching, and Tracing

The latest PDF identifies the trap precisely: a proposal about representation gets translated too quickly into a proposal about drawing. Then the cone becomes rays, the one-bit state becomes pixels, $B = 0$ becomes a sampled volume, and grammar becomes mesh generation. The report argues that this translation can be useful yet still erase the intended inversion. That paragraph is the cleanest internal statement of how the proposed substrate differs from the familiar pipelines. [filecite turn0file0L24-L29](#)

The comparison becomes clearer when framed as “what question is the system trying to answer?”

System	Basic question it answers	Why it overlaps with the corpus	Why it is not the corpus’s core idea
Rasterization	How do stored surfaces project into pixels this frame?	The corpus can still emit images as downstream projections	The latest PDF explicitly rejects taking the projection as the world itself

System	Basic question it answers	Why it overlaps with the corpus	Why it is not the corpus's core idea
Raymarching	What does repeated field sampling along projected directions reveal?	The corpus often uses SDF language and cone imagery that tempt this reading	The mature concept replaces repeated sampling with compatibility, event solving, and transition semantics
Ray tracing	Which transport or visibility paths connect a point, source, or eye through space?	Cones, visibility, and support domains can later be rendered or traced	The latest PDF redefines cones as causal, semantic, collision, memory, and grammar supports, not just light transport
Proposed substrate	What state holds at time t , what event happens next, and which sectors can couple?	Uses implicit geometry, supports, invariants, and local charts that can be projected if desired	Treats representation as a directly queryable relation algebra and event calculus

The underlying inversion appears in the latest PDF's "Default pipeline question / Equation-world question" table. Traditional questions ask what geometry must be stored, what updates every frame, and how sensors obtain images. The equation-world version asks what finite relations generate possible states, what the state is at time t , what the next event is after t_0 , and what events fall inside an analytic domain of dependence. That is not a minor implementation tweak. It is a shift in ontology.

[filecite\turn0file0\101-L118](#)

This also explains the overlap. The corpus does **not** require hostility to rasterization, raymarching, or tracing as techniques. It only refuses to let them define the substrate. The latest PDF explicitly says those can remain downstream projection tools. In that sense, the relationship is not "mutual exclusion." It is "do not mistake the projector for the state space." [filecite\turn0file0\120-L23](#)

The archive itself illustrates both the confusion and the correction. The posterize/aerosol/lighthouse material starts close to a shader or volumetric effect framing, and parts of *StepsWithout0* still imagine the breakthrough as "getting past zero" without sampling. But the mid-July architectural reports already push back, saying the cone should be treated as a sensing or support operator over a shared manifold, not as a rendering trick; and the newest PDF completes that correction by naming cones "analytic relevance operators" that unify causality, collision, memory retrieval, and procedural unfolding without turning them into visual rays. [filecite\turn0file0\1260-L300](#) (posterize-shader...aerosol-spray-in-a-cone...lighthouse.pdf, pp. 1–5, 29–33) (*StepsWithout0*.pdf, pp. 22, 43–45) (Aerosol Cones, SDFs, and Matryoshka Nesting in the Unified Engine.pdf, pp. 1–2, 5)

So, on the user's central concern: yes, the corpus's through line is fundamentally different from rasterization and raymarching **when read at the level the latest PDF demands**. The difference lies less in sphere-vs-grid geometry and more in **query-vs-projection semantics**. Rasterization and raymarching ask how to sample or draw. The spherical substrate line asks how to organize support, compatibility, and transitions so that state and event can be queried directly. [filecite\turn0file0\195-L118](#) [filecite\turn0file0\194-L197](#)

Risks and Experiments

The latest PDF's bottleneck list is the most valuable research agenda in the corpus because it names the places where the idea could fail **without** falling back into a graphics benchmark. The real hazards are algebraic closure, event explosion, branch explosion, degeneracies, conditioning, exogenous entropy, identity under split/merge, self-reference, semantic grounding, and query governance. Those are exactly the right risks for this kind of architecture. [\[turn0file0L455-L470\]](#)

From those bottlenecks, the most important open questions become clear.

First, can grammar composition stay inside a closed or normalized relation family, or does the expression size grow until direct query loses any advantage? That is the core algebra question. Second, can compatibility predicates remove enough candidate events to keep solving sparse? That is the scaling question. Third, can lineage survive splits, merges, and reconciliations without devolving into ad hoc bookkeeping? That is the identity question. Fourth, can the shared world/agent ontology carry actual semantics rather than just colocated tensors and phase tags? That is the cognitive question. And fifth, can all of that remain numerically stable in the presence of singularities and long horizons? That is the implementation question. [\[turn0file0L368-L375\]](#) [\[turn0file0L382-L450\]](#) [\[turn0file0L455-L470\]](#)

The archive already contains a strong prototype program. The first phase should revolve around six required queries: state at time, next event, events in cone, phase coupling of co-located sheets, transition routing through the hourglass or gluing rule, and persistence/identity reconstruction from seed plus external log. Those six queries are the right minimal experiment because together they test directness, support semantics, compatibility, routing, lineage, and memory discipline. [\[turn0file0L492-L503\]](#)

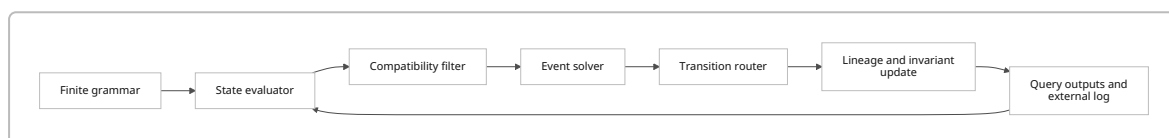
A disciplined first-wave experiment set would look like this:

Prototype experiment	What it tests	Evidence of success
Horizon skipping	Whether <code>state_at(t)</code> cost scales with skipped time or only with expression complexity	Query cost stays tied to expression size, not elapsed steps
Co-located phase sheets	Whether double vacuum is a real compatibility sector and not just duplicated coordinates	Same- x states remain uncoupled until explicit compatibility is met
Event routing at $B = 0$	Whether transitions preserve invariants while routing across sheets or chambers	Correct event ordering and stable invariant tags
Grammar depth stress	Whether direct-query advantages survive rule composition	Growth stays bounded or normalizable within the allowed relation family
Identity split/merge	Whether generative address can survive nontrivial persistence operations	Lineage collisions are detectable and reconstructable

Prototype experiment	What it tests	Evidence of success
Minimal agent integration	Whether action, memory, and perception can reuse the same relation substrate	Agent queries and action selection operate in the same transition language as embodiment

Those experiments are not speculative additions by me; they directly condense the latest PDF's own evaluation program and exit criteria. [filecite\[turn0file0L512-L559\]](#)

A clean prototype flow looks like this:



The most important recommendation for a proof of concept is therefore to **refuse two temptations**. The first temptation is to turn the experiment immediately into a renderer benchmark. The second temptation is to preserve the archive's most inflated claims as requirements. The proper goal is narrower: show that one nontrivial relation family can answer state-at-time and next-event queries without stepping every frame, under explicit compatibility, with stable lineage, bounded expression growth, and memory proportional to novelty rather than to geometric extent. If that works, the prototype has validated the core of the spherical substrate line. If it fails, the failure will be informative in exactly the right place. [filecite\[turn0file0L544-L559\]](#)

The corpus's own final decision is the correct closing judgment: **GO for formalization and Equation World Zero, but do not begin by recreating a conventional scene or image pipeline. Begin with the algebra of state, compatibility, event, transition, identity, and memory.** [filecite\[turn0file0L588-L592\]](#)